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NIM = 2519019029
Kelas = A

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2.) $A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix}$; $A = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 4 & 2 \\ 2 & 1 & 5 \end{bmatrix}$

a.) Rumus : $\det(A) = a_{11}(C_{11}) + a_{12}(C_{12}) + a_{13}(C_{13})$

•) Elemen $a_{11} = 3$:

$\det \begin{vmatrix} 4 & 2 \\ 1 & 5 \end{vmatrix} = (4 \cdot 5) - (2 \cdot 1) = 20 - 2 = 18$

•) Elemen $a_{12} = 2$

$-\det \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = (1 \cdot 5) - (2 \cdot 2) = 5 - 4 = 1$

•) Elemen $a_{13} = 2$

$\det \begin{vmatrix} 1 & 4 \\ 2 & 1 \end{vmatrix} = (1 \cdot 1) - (4 \cdot 2) = 1 - 8 = -7$

$\det(A) = 3(18) + 2(-1) + 2(-7)$

$\det(A) = 54 - 2 - 14 = 38$ //

b.) (M_{ij}) dan kofaktor

•) $M_{11} = 18$, $M_{12} = 1$, $M_{13} = -7$

•) $M_{21} = \det \begin{vmatrix} 2 & 2 \\ 1 & 5 \end{vmatrix} = 10 - 2 = 8$

•) $M_{22} = \det \begin{vmatrix} 3 & 2 \\ 2 & 5 \end{vmatrix} = 15 - 4 = 11$

•) $M_{23} = \det \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = 3 - 4 = -1$

•) $M_{31} = \det \begin{vmatrix} 2 & 2 \\ 4 & 2 \end{vmatrix} = 4 - 8 = -4$

•) $M_{32} = \det \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} = 6 - 2 = 4$

•) $M_{33} = \det \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = 12 - 2 = 10$

$C = \begin{bmatrix} 18 & -1 & -7 \\ -8 & 11 & 1 \\ -4 & -4 & 10 \end{bmatrix}$

c.) Matriks Adjoin dan Invers (A^{-1})

$Adj(A) = C^T = \begin{bmatrix} 18 & -8 & -4 \\ -1 & 11 & -4 \\ -7 & 1 & 10 \end{bmatrix}$

Invers A^{-1} : $A^{-1} = \frac{1}{\det(A)} \cdot Adj(A) = \frac{1}{38} \cdot \begin{bmatrix} 18 & -8 & -4 \\ -1 & 11 & -4 \\ -7 & 1 & 10 \end{bmatrix}$

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c.) $P = \begin{bmatrix} 120 & 50+29 & 200 \\ 80 & 2+2 & 10+29 \\ 30+29 & 90 & 170 \end{bmatrix} = \begin{bmatrix} 120 & 79 & 200 \\ 80 & 4 & 39 \\ 59 & 90 & 170 \end{bmatrix}$

Enkripsi $E = A \cdot P$

$A = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 4 & 2 \\ 2 & 1 & 5 \end{bmatrix} \cdot P = \begin{bmatrix} 120 & 79 & 200 \\ 80 & 4 & 39 \\ 59 & 90 & 170 \end{bmatrix}$

Baris 1 : $E_{11} = (3 \cdot 120) + (2 \cdot 80) + (2 \cdot 59) = 360 + 160 + 108 = 628$

$E_{12} = (3 \cdot 79) + (2 \cdot 4) + (2 \cdot 90) = 222 + 8 + 180 = 410$

$E_{13} = (3 \cdot 200) + (2 \cdot 39) + (2 \cdot 170) = 600 + 68 + 340 = 1008$

Baris 2 : $E_{21} = (1 \cdot 120) + (4 \cdot 80) + (2 \cdot 59) = 120 + 320 + 108 = 548$

$E_{22} = (1 \cdot 79) + (4 \cdot 4) + (2 \cdot 90) = 79 + 16 + 180 = 270$

$E_{23} = (1 \cdot 200) + (4 \cdot 39) + (2 \cdot 170) = 200 + 136 + 340 = 676$

Baris 3 : $E_{31} = (2 \cdot 120) + (1 \cdot 80) + (5 \cdot 59) = 240 + 80 + 270 = 590$

$E_{32} = (2 \cdot 79) + (1 \cdot 4) + (5 \cdot 90) = 158 + 4 + 450 = 602$

$E_{33} = (2 \cdot 200) + (1 \cdot 39) + (5 \cdot 170) = 400 + 39 + 850 = 1289$

Hasil Matriks Enkripsi E :

$E = \begin{bmatrix} 628 & 410 & 1008 \\ 548 & 270 & 676 \\ 590 & 602 & 1289 \end{bmatrix}$

* Membuktikan Deskripsi $P' = A^{-1} \cdot E$

$P'_{11} = \frac{1}{38}[(18 \cdot 628) + (-8 \cdot 548) + (-4 \cdot 590)]$

$P'_{11} = \frac{1}{38}[11304 - 4384 - 2360]$

$P'_{11} = \frac{1}{38}[4560] = 120$

Apa yang terjadi jika det(A) = 0 Bermasalah ?

- Matriks Singular
- Dalam proses Deskripsi, kita membutuhkan A^{-1}. jika A^{-1} tidak ada maka tidak bisa membalikkan proses enkripsi

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